

# Martin Nunction

Martin Nunction(a pun for Notation/Function), Is a large number notation created by Martin Wanralfs or UpBeatSquirrel925 in Fandom wiki.

## A.Definitions

### 1.Martin Aunction (MA)

Martin Aunction is the very first function in Martin nunction, written as:

$$A(a,b,c)$$

Let a be any non-zero finite positive integer, a is the argument, a is the first entry.

Let b be any non-zero finite positive integer, b is the amount of (c-th) hyperoperation, b is the second entry.

Let c be any non-zero finite positive integer, c is the hyperoperation level, c is the third entry.

$$A(a) = a$$

$$A(a,b) = a^b$$

General Martin Aunction expansion:

$$A(a,b,c) = a (c)^b a = a (c)^{b-1} (a (c)^{b-1} (a (c)^{b-1} \dots \#.$$

Right associatively, using right operand a value as the amount of iteration.

(Note: octothorpe symbol means unchanged remainder of an array and hyperoperation.)

$$A(a,b,c) = a (c\text{-th level hyperoperation})^b a.$$

General Martin Aunction leveling down:

$$A(a,1,c) = a (c)^1 a = a (c-1)^a a, \text{ or}$$

$$A(c)^1 n = a (c-1)^n a.$$

Where b = 1, b value replaced by right operand a, while c is subtracted by 1, moving to (c-1)-th hyperoperation.

Examples:

$$A(1,b,c) = 1 (c)^b 1 = 1$$

$$A(1,1,1) = 1 \uparrow^1 1 = 1^1 = 1$$

$$A(1,n,n) = 1 (n)^n 1 = 1$$

Where  $c = 1$  is the Knuth's up arrow hyperoperation ( $\uparrow^b$ ),  $a = 1$ , left  $a$  is the argument while right  $a$  is hyperoperation expansion operand, and  $b = 1$  is the amount of current level hyperoperation,  $a$  becomes the result ignoring the second and third entry iff  $a = 1$ .

$$A(2,2,2) = 2 (2)^2 2 = 2 (2)^{12} = 2 \uparrow^2 2 = 2 \uparrow 2 = 2^2 = 4.$$

$$A(2,b,c) = 2 (c)^b 2 = 2 (c)^{b-1} 2 = 2 (c)^{b-2} 2 \dots$$

Operation will terminate to 4 such that  $a = 2$ .

$$A(3,2,1) = 3 \uparrow^2 3 = 3 \uparrow (3 \uparrow 3) = 3 \uparrow 27 = 7,625,597,484,987.$$

## 2. Martin Array Aunction (MAA, Level 1)

The next level in Martin Nunction, extending the array length of Martin Aunction, written as

$$A(a,b,c,d,e,\#)$$

(Note: octothorpe symbol means unchanged remainder of an array and hyperoperation.)

Where array entry must be a non-zero finite positive integer or an array.

General 4 Entry Expansion:

$$A(ab,c,d) = A(a,b,(a,b-1,c,d),d-1)$$

General Pairing rule for Entry  $>3$  where there's a number 1 followed by a number 2, and where 1 followed by a number  $> 2$ , scanning right to left:

$$A(a,1,2,\#) = A(a,3,1,\#)$$

$$A(a,1,c,\#) = A(a,(a,(\dots\#\dots)),c-1,\#),c-1,\#)$$

Where  $b$  is now replaced with an entire array nested into  $b$  entry  $c$  times, subtracting  $c$  by 1 on the outer array and all inner arrays

General  $n$  entries:

$$A(a,b,c,d,\#) \rightarrow A(a,b,c,1,\#), \text{ Entries after } d \text{ are for applying pairing rule such that any entry except for } a = 1.$$

Example:

$$A(3,2,2,2) = A(3,2,(3,1,2,2),1) \rightarrow A(3,2,(3,1,2,2)), \text{ remove trailing } 1, \text{ according to rule 3.}$$

→ Applying pairing rule, for pair  $b=1$  and  $c=2$ ,  $(1,2)$ .

$$A(3,2,(3,3,1,2))$$

→ Applying pairing rule, for pair  $c=1$  and  $d=2$ ,  $(1,2)$ .

$$A(3,2,(3,3,3,1)) \rightarrow A(3,2,(3,3,3)), \text{ remove trailing } 1.$$

→ Let  $X = A(3,3,3)$

$$\rightarrow A(3,2,X) = 3 (X)^2 3.$$

### 3. Martin Array Nunction (MAN, base array universe)

Martin Array Nunction(MAN)

$A_{-}(a)(n,l)$ .

let (a) = array level

let n = value of entries in current array level

let l = length or amount of entries in current array level

$A_{-}(1)(n,4) = A(n,n,n,n)$

$A_{-}(2)(n,3) = B(n,n,n)$

General Leveling down when l = 1 or 2 or 3:

$A_{-}(n)(n,1) = n(n) = n$

→  $A_{-}(n-1)(n,n)$

$A_{-}(n)(n,2) = n(n,n) = n^n$ , let the result be X

→  $A_{-}(n-1)(n,X)$

$A_{-}(n)(n,3) = n(n,n,n) = n$  (n-th hyperoperation) $^n n$ , let the result be X

→  $A_{-}(n-1)(n,X)$ .

Replace parameter l with result of (n) level array calculation

Then subtract (n) by 1.

General l = 4 expansion for any (a):

$A_{-}(a)(n,4) = a(n,n,n,n) = a(n,n,(n,n-1,n,n),n-1)$

Array with l > 4 or more than 4 entries are only for applying the pairing rule.

General > 1 entry parameter (a):

$A_{-}(a,b)(n,l) = \text{Let } X = A_{-}(a)(n,l)$

→  $A_{-}(X,b-1)(n,l)$

$A_{-}(a,b,\#j,k)(n,l) = \text{Let } X = A_{-}(a,b,\#j)(n,l)$

→  $A_{-}(X,X,\#X,k-1)(n,l)$

Calculate (a) parameter with one lower entry length before k-th entry.

Then Replace every entry before k with X and subtract k-th entry by 1.

#### 4. Bangs (!)

An Array operator of MAN, the first priority Operator, written in the operator slots which are second to last entry in MAN a parameter entry.

$$A_{\#}(\text{operator}, k)(n, l) \text{ or } A_{(a, !, b)}(n, l, !)$$

General entry  $\leq 3$  with single or more bangs:

$$A_{(a, !, b)}(n, l) = A_{(a, a, \#)}(n, l)$$

a parameter with  $L_a(a, a)$  entries.

$$A_{(a, b, !, c)}(n, l) = A_{(a, a, \#)}(n, l)$$

a parameter array with  $A_{(a, !, b)}(a, a)$  entries.

General MAN a parameter array with n bangs:

$$A_{(a, !^n, b)}(n, l) = A_{(a, a, \dots \#, !^{n-1}, a)}(n, l)$$

MAN current array filled with a with  $A_a(a, a)$  Entries.

$$A_{(a, b, !^n, c)}(n, l) = A_{(a, a, \dots \#, !^{n-1}, a)}(n, l) =$$

MAN a parameter array with (n-1) bangs and  $L_{(a, !^n, b)}(a, a)$  entries.

>3 entry with bangs recursion:

$$A_{(a, b, c, !, d)}(n, l) = \text{Let } X = A_{(a, b, !, c)}(n, l), \text{ calculate 3 Entry a parameter with single bang first.}$$

$$\rightarrow A_{(X, X, X, !, d-1)}(X, X)$$

General with n bangs:

$$A_{(a, b, \#, j, !^n, k)}(n, l) = \text{Let } X = A_{(a, b, \#, !^n, j)}, \text{ calculate a parameter with one entry lower with n bangs first.}$$

$$\rightarrow A_{(X, X, \#, X!^{\uparrow(n)}, k-1)}(X, X)$$

Replace every entry before bang with X, along with n and l parameters.

#### 5. Tiers (/n/)

The second priority operator.

$$A_{(a, /n/, b)}.$$

General Tiering:

/n/ = Tier 1 with value n, //n// = Tier 2 with value n, (k-th tier(n) = k-th tier with value n, higher tier = less priority.

General tier 1 with  $n = 1$  and  $n > 2$ :

$$A_{\text{a,/1/,b}}(n,l) = \text{Let } X = A_{\text{a,b}}(n,l), \rightarrow A_{\text{a,!^((L_{\text{#}})),b}}(n,l)$$

Evaluate X, then subtract tier 1 Value by 1, which becomes X amounts of Bangs, with entire function nested into bang exponent, where innermost nested function =  $A_{\text{a,!^X,b}}(n,l)$

$$A_{\text{a,/n/,b}}(n,l) = \text{Let } X = A_{\text{a,b}}(n,l)$$

$$\rightarrow A_{\text{a,/n-1/,!^X(A_{\text{#}})),b}}(n,l)$$

Evaluate X, then subtract tier 1 value by 1, which becomes X amounts of Bangs added to the second rightmost entry as higher priority operator, with entire function nested into it's exponent, Where the innermost nested function =  $A_{\text{a,/n-1/,!^X,b}}(n,l)$

General Tier 2 and Tier k with value n:

$$A_{\text{a,//n//,b}}(n,l) = \text{Let } X = A_{\text{a,b}}(n,l)$$

$$\rightarrow A_{\text{a,//n-1//,A_{\text{#}}/b}}(n,l)$$

$$\text{Where innermost nested function} = A_{\text{a,//n-1//,X/b}}(n,l)$$

Nest entire function X times inside Tier 1 bracket and subtract Tier 2 value by 1, then add lower tier operator on the right side of the higher tier operator in the operator slots

$$A_{\text{n,(k-tier(n)),b}}(n,l) = \text{let } X = A_{\text{a,b}}(n,l)$$

$$\rightarrow A_{\text{n,(k-tier(n-1)),(k-1 tier(),b)}}(n,l)$$

Where innermost nested function =

$$A_{\text{a,(k-tier(n-1)),(k-1 tier(X),b)}}(n,l)$$

Nest entire function into (k-1)-th tier bracket and subtract k-th tier value by 1, then add lower tier operator on the right side of the higher tier in the operator slots.

## 6.Martin Kunction (MK)

Index of Structure and tiers, written as:

$$M_{\text{k(m,n,l)}}$$

Let k = k-th tier structure within Martin Nunction.

Let m = tier within Current tier structure

Let n = amount of current tier within the structure

Let l = A\_(l,l,#)(l,l), with l entries for a parameter.

M\_k(m,n,3) = A\_(3,3, (k-th structure tier m(n)), 3)(3,3)

Example:

M\_1(1,1,3) = A\_(3,3,/1/,3)(3,3)

M\_1(2,1,3) = A\_(3,3,//1//,3)(3,3)

M\_2(1,2,2) = A\_(2,{2},2)(2,2)

General Martin Kunction:

M\_k(m,n,l) = A\_(#, (k-structure tier m(n)), l)(l,l) = Let X = A\_(#,l)(l,l)

→ A\_(#, (k-structure tier m(n-1)), (k-structure tier m-1(A\_(#)), l)(l,l)

Calculate X, subtract k-th structure m-th tier value by 1, then add k-th Structure (m-1)-th tier with value X, nest entire function into k-th sturture (m-1)-th tier bracket.

Where Innermost nested functikn = A\_(#, (k-structure m(n-1)), (k-structure tier m-1(X)), l)(l,l).

Leveling Down Structure:

A\_(#, (k-structure tier 1(1)), l)(l,l) = Let X = A\_(#,l)(l,l)

→ A\_(#,l, (k-1 structure tier X(A\_(#)), l)(l,l)

Calculate X, Level down to (k-1)-th structure with tier X and Value X, nest the entire function into the (k-1)-th structure X-th tier bracket X times, with Innermost nested function = A\_(#, (k-1 Structure tier X(X)), l)(l,l)

## 7.Martin Extended Array Nunction(MEAN)

A^v(k,n,l)

Let v = array universe

Let k = k-th structure k-th tier value k operator. Such that l > 1.

Ignore k if l = 1.

Let n = value of each entries

Let l = length.

$$L^1(n,n,1) = A_{-}(n)(n,n) = n(n,n\#), \text{ base Universe}$$

$$L^1(n,n,2) = A_{-}(n,(n\text{-th structure tier } n(n)),n)(n,n)$$

$$L^2(1,n,1) = A^1_{-}(n)(n,n) = n_1(n,n\#)$$

General:

$$L^n(n,n,n) = A^{n-1}_{-}(\#, (n\text{-th structure tier } n(n)),n)(n,n)$$

Calculating higher universe Martin Array Auction:

$A(a,b\#)$  = Martin Array Auction, Base Universe.

$A_1(a,b,\#)$  = Martin Array Auction, Universe 1.

$n(a,b,\#)$  = MAA level  $n$ , base Universe

$n_1(a,b,\#)$  = MAA level  $n$ , universe 1.

$$A_1(a,b,c) = a \text{ (c-th hyperoperation level)}^b a$$

Any  $v$  MA or MAA has the same mechanism as  $v = 1$ .

Leveling down universe:

$$L^2(n,n,1) = L^1_{-}(n)(n,n) = n_1(n,n\#), \text{ Let } X = n_1(n,n,\#)$$

$$\rightarrow L^1(X,X,X).$$

Example:

$$L^2(n,2,1) = L^1_{-}(2)(2,2) = L^1_{-}(1)(2,(L^1_{-}(1)(2,2))).$$

$$\text{Calculate } L^1_{-}(1)(2,2) \text{ first} = A_1(2,2) = 2^2 = 4$$

$$\begin{aligned} \rightarrow L^1_{-}(1)(2,4) &= A_1(2,2,2,2) = A_1(2,2,(2,1,2,2),1) \rightarrow (2,2,(2,1,2,2)) \rightarrow (2,2,(2,3,1,2)) \rightarrow (2,2,(2,3,3,1)) \\ &\rightarrow (2,2,(2,3,3)) \rightarrow (2,2,4) = 4. \end{aligned}$$

$$\rightarrow L^1(4,4,4) = L_{-}(4,4,(4\text{-th structure tier } 4(4)),4)(4,4)$$

## B.Rules

Rule 1: MAN array or function entry must contain an array, function, or non zero positive integers, except for operator slots.

Rule 2:  $(\#, \text{Operators}, k)$  or  $L\_a(n, l, \text{operators}), b$  or  $(a, [n], \{n\}, /n/, !, b)$  or  $L\_a(n, l, \{n\}, /n/, !)$ . Higher priority operator is written right before last entry, least Priority written after Second last array entry and so on, operator slots are always second to rightmost entry in MAN.

Rule 3: remove trailing 1,  $(\#, 1) \rightarrow (\#)$ , or  $(\#, j, !, 1) \rightarrow (\#, !, j)$

Rule 4: Prioritize (!) before m-th tier, then k-th structure.

Rule 5: Evaluate Innermost array/nested function first.

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